OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

12.5 Cross paths

Cross paths are similar to crossings from the driving point of view, adding connections between specific lanes. Cross paths are part of the junction similar to connecting roads.

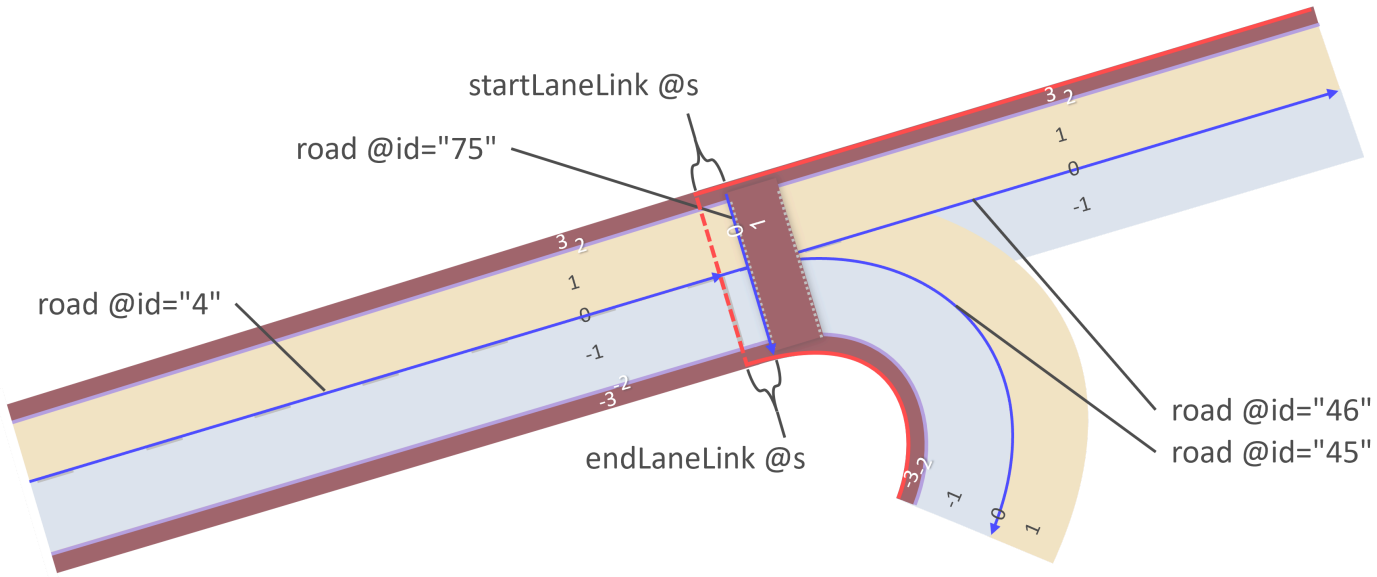

Figure 85. Example of a cross path with a pedestrian crossing in a common junction

Figure 85 shows a part of a common junction where the road with @id="75" as cross path connects the sidewalks of the two connecting roads with @id="45" and @id="46". The @roadAtStart attribute of the `<crossPath>` element defines which road the `<startLaneLink>` element refers to with its s@ attribute. The @roadAtEnd attribute of the `<crossPath>` element defines which road the `<endLaneLink>` element refers to with its s@ attribute.

Cross paths are represented by `<crossPath>` elements within the `<junction>` element. A `<crossPath>` element defines the crossing road with the value of the @crossingRoad attribute and the roads connected by the cross path with the values of the @roadAtStart and @roadAtEnd attributes.

Elements in UML model

`<crossPath>` element

In ASAM OpenDRIVE, cross paths are represented by `<crossPath>` elements within the `<junction>` element.

```
UML class:  t_junction_crossPath
XML tag:    <crossPath> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Cross paths are intended for pedestrian crossings and are junctions elements where traffic of a lane can cross other lanes and continue on a different lane of the same or a different road. The cross path itself is a separate road.

Table 64. Attributes of the <crossPath> element

Name	Type	Use	Introduced	Description
`crossingRoad`	string	required	1.8.0	ID of road defining the cross path.
`id`	string	required	1.8.0	Unique ID within the junction
`roadAtEnd`	string	required	1.8.0	ID of road at end point of the crossing road
`roadAtStart`	string	required	1.8.0	ID of road at start point of the crossing road

`<startLaneLink>` and `<endLaneLink>` elements

In ASAM OpenDRIVE, lane links of cross paths are represented by `<startLaneLink>` and `<endLaneLink>` elements within the `<crossPath>` element.

```
UML class:  t_junction_crossPath_laneLink
XML tag:    <endLaneLink> (Multiplicity: 1)
XML tag:    <startLaneLink> (Multiplicity: 1)
Introduced: 1.8.0
```

Define the links between the lanes of the `<crossPath>` to the lanes of other roads.

Table 65. Attributes of the <endLaneLink> and <startLaneLink> elements

Name	Type	Use	Introduced	Description
`from`	integer	required	1.8.0	Lane ID of either @roadAtEnd for `<endLaneLink>` or @roadAtStart for `<startLaneLink>`
`s`	t_grEqZero	required	1.8.0	s-coordinate of either start or end point in linked road.
`to`	integer	required	1.8.0	Lane ID of @crossingRoad

XML example

```xml
<road name="" length="6.8257399999999990e+01" id="4" junction="-1" rule="RHT">
    <link>
        <successor elementType="junction" elementId="10"/>
    </link>
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="3" type="walking" level="false">
                    <link/>
                    ...
                </lane>
                <lane id="2" type="curb" level="false">
                    <link/>
                    ...
                </lane>
                <lane id="1" type="driving" level="false">
                    <link/>
                    ...
                </lane>
            </left>
            <center>
                <lane id="0">
                    ...
                </lane>
            </center>
            <right>
                <lane id="-1" type="driving" level="false">
                    <link/>
                    ...
                </lane>
                <lane id="-2" type="curb" level="false">
                    <link/>
                    ...
                </lane>
                <lane id="-3" type="walking" level="false">
```

```
                    <link/>
                    ...
                </lane>
            </right>
        </laneSection>
    </lanes>
    ...
</road>
<road name="" length="3.9995576531488346e+01" id="46" junction="10" rule="RHT">
    <link>
        <predecessor elementType="road" elementId="4" contactPoint="end"/>
        <successor elementType="road" elementId="6" contactPoint="end"/>
    </link>
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="3" type="walking" level="false">
                    <link>
                        <predecessor id="3"/>
                        <successor id="-3"/>
                    </link>
                    ...
                </lane>
                <lane id="2" type="curb" level="false">
                    <link>
                        <predecessor id="2"/>
                        <successor id="-2"/>
                    </link>
                    ...
                </lane>
                <lane id="1" type="driving" level="false">
                    <link>
                        <predecessor id="1"/>
                        <successor id="-1"/>
                    </link>
                    ...
                </lane>
```

```
4               </left>
2               <center>
4                   <lane id="0">
3                       ...
4                   </lane>
4               </center>
4               <right>
5                   <lane id="-1" type="driving" level="false">
4                       <link>
6                           <predecessor id="-1"/>
4                           <successor id="1"/>
7                       </link>
4                       ...
8                   </lane>
4               </right>
9           </laneSection>
5       </lanes>
0   </road>
5   <road name="" length="2.8900504898771064e+01" id="45" junction="10" rule="RHT">
1       <link>
5           <predecessor elementType="road" elementId="4" contactPoint="end"/>
2           <successor elementType="road" elementId="5" contactPoint="start"/>
5       </link>
3       ...
5       <lanes>
4           <laneSection s="0.0000000000000000e+00">
5               <left>
5                   <lane id="1" type="driving" level="false">
5                       <link>
6                           <predecessor id="1"/>
5                           <successor id="1"/>
7                       </link>
5                       ...
8                   </lane>
5               </left>
9               <center>
6                   <lane id="0">
```

```
                    ...
                </lane>
            </center>
            <right>
                <lane id="-1" type="driving" level="false">
                    <link>
                        <predecessor id="-1"/>
                        <successor id="-1"/>
                    </link>
                    ...
                </lane>
                <lane id="-2" type="curb" level="false">
                    <link>
                        <predecessor id="-2"/>
                        <successor id="-2"/>
                    </link>
                    ...
                </lane>
                <lane id="-3" type="walking" level="false">
                    <link>
                        <predecessor id="-3"/>
                        <successor id="-3"/>
                    </link>
                    ...
                </lane>
            </right>
        </laneSection>
    </lanes>
    ....
</road>
<road name="" length="9.4332512961489723e+00" id="75" junction="10" rule="RHT">
    <link/>
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="1" type="walking" level="false">
```

```
                    <link/>
                    ...
                </lane>
            </left>
            <center>
                <lane id="0">
                    ...
                </lane>
            </center>
        </laneSection>
    </lanes>
    ...
</road>
<junction name="" id="10">
    <connection id="0" incomingRoad="4" connectingRoad="46" contactPoint="start">
        <laneLink from="-1" to="-1"/>
    </connection>
    <connection id="1" incomingRoad="4" connectingRoad="45" contactPoint="start">
        <laneLink from="-1" to="-1"/>
        <laneLink from="-2" to="-2"/>
        <laneLink from="-3" to="-3"/>
    </connection>
    ...
    <crossPath id="6" crossingRoad="75" roadAtStart="46" roadAtEnd="45">
        <startLaneLink s="5.0000000000000000e-01" from="3" to="1"/>
        <endLaneLink s="2.4841630000000000e-01" from="-3" to="1"/>
    </crossPath>
    ...
</junction>
```

```
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```

```
7
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7
4
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7
5
1
7
6
```

Rules

The following rules apply to cross paths:

- asam.net:xodr:1.8.0:junctions.cross_path.within_junction_area: Cross paths shall be within the area of a common junction or a virtual junction.
- asam.net:xodr:1.8.0:junctions.cross_path.disregard_cross_road_evelation: The elevations of the crossing road defined by the @crossingRoad attribute of the `<crossPath>` element are disregarded.
- asam.net:xodr:1.8.0:junctions.cross_path.lane_linkage: Start and end of the crossing road shall reach the linked lanes specified by the `<startLaneLink>` and `<endLaneLink>` elements.
- asam.net:xodr:1.8.0:junctions.cross_path.start_end_contained: The start and end points of the crossing road and its lanes shall be fully contained within the linked lanes specified by the `<startLaneLink>` and `<endLaneLink>` elements.
- asam.net:xodr:1.8.0:junctions.cross_path.only_connect_correct_type: Cross paths shall only connect lanes with @type="walking" or @type="biking".

- asam.net:xodr:1.8.0:junctions.cross_path.correct_junction_id: The @junction attribute shall contain the id of the junction to which a road belongs.

Related topics

- Section 12.7.1, "Cross paths with virtual junctions"
- Section 12.8, "Crossings"